

Subroutine ODEX

Extrapolation code for $y' = f(x, y)$, based on the GBS algorithm (Section II.9). It uses variable order and variable step sizes and is provided with a high-order dense output. Again, the missing parts in the description are identical to those of DOPRI5.

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      SUBROUTINE ODEX(N,FCN,X,Y,XEND,H,
+          RTOL,ATOL,ITOL,
+          SOLOUT,IOUT,
+          WORK,LWORK,IWORK,LIWORK,LRCNT,LICNT,IDID)
C -----
C   NUMERICAL SOLUTION OF A SYSTEM OF FIRST ORDER
C   ORDINARY DIFFERENTIAL EQUATIONS  Y'=F(X,Y).
C   THIS IS AN EXTRAPOLATION-ALGORITHM (GBS), BASED ON THE
C   EXPLICIT MIDPOINT RULE (WITH STEPSIZE CONTROL,
C   ORDER SELECTION AND DENSE OUTPUT).
C
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C            DENSE OUTPUT WRITTEN BY E. HAIRER AND A. OSTERMANN
C
C   .....
C
C   VERSION DECEMBER 18, 1991
C   .....
C
C   H          INITIAL STEP SIZE GUESS;
C              H=1.DO/(NORM OF F'), USUALLY 1.D-1 OR 1.D-3, IS GOOD.
C              THIS CHOICE IS NOT VERY IMPORTANT, THE CODE QUICKLY
C              ADAPTS ITS STEP SIZE. WHEN YOU ARE NOT SURE, THEN
C              STUDY THE CHOSEN VALUES FOR A FEW
C              STEPS IN SUBROUTINE "SOLOUT".
C              (IF H=0.DO, THE CODE PUTS H=1.D-4).
C
C   .....
C
C   ----- CONTINUOUS OUTPUT (IF IOUT=2): -----
C           DURING CALLS TO "SOLOUT", A CONTINUOUS SOLUTION
C           FOR THE INTERVAL [XOLD,X] IS AVAILABLE THROUGH
C           THE REAL*8 FUNCTION
C               >>> CONTEX(I,S) <<<
C           WHICH PROVIDES AN APPROXIMATION TO THE I-TH
C           COMPONENT OF THE SOLUTION AT THE POINT S. THE VALUE
C           S SHOULD LIE IN THE INTERVAL [XOLD,X].
C
C   .....
C
C   WORK       ARRAY OF WORKING SPACE OF LENGTH "LWORK".
C              SERVES AS WORKING SPACE FOR ALL VECTORS.
C              "LWORK" MUST BE AT LEAST
C              N*(KM+5)+5*KM+10+2*KM*(KM+1)*NRDENS
C              WHERE NRDENS=IWORK(8) (SEE BELOW) AND
C              KM=9          IF IWORK(2)=0
C              KM=IWORK(2)   IF IWORK(2).GT.0
C              WORK(1),...,WORK(10) SERVE AS PARAMETERS
C              FOR THE CODE. FOR STANDARD USE, SET THESE
C              PARAMETERS TO ZERO BEFORE CALLING.
C
C   .....

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C
C      IWORK      INTEGER WORKING SPACE OF LENGTH "LIWORK".
C                  "LIWORK" MUST BE AT LEAST
C                      2*KM+10+NRDENS
C                  IWORK(1),...,IWORK(9) SERVE AS PARAMETERS
C                  FOR THE CODE. FOR STANDARD USE, SET THESE
C                  PARAMETERS TO ZERO BEFORE CALLING.
C
C      .....
C      LRCONT     DECLARED LENGTH OF COMMON BLOCK
C                  >>> COMMON /CONTR/RCONT(LRCONT) <<<
C                  WHICH MUST BE DECLARED IN THE CALLING PROGRAM.
C                  "LRCONT" MUST BE AT LEAST
C                      ( 2 * KM + 5 ) * NRDENS + 2
C                  WHERE KM=IWORK(2) AND NRDENS=IWORK(8) (SEE BELOW).
C
C      LICONT     DECLARED LENGTH OF COMMON BLOCK
C                  >>> COMMON /CONTI/ICONT(LICONT) <<<
C                  WHICH MUST BE DECLARED IN THE CALLING PROGRAM.
C                  "LICONT" MUST BE AT LEAST
C                      NRDENS + 2
C                  THESE COMMON BLOCKS ARE USED FOR STORING THE COEFFICIENTS
C                  OF THE CONTINUOUS SOLUTION AND MAKES THE CALLING LIST FOR
C                  THE FUNCTION "CONTEX" AS SIMPLE AS POSSIBLE.
C
C      .....
C      WORK(2)    MAXIMAL STEP SIZE, DEFAULT XEND-X.
C
C      WORK(3)    STEP SIZE IS REDUCED BY FACTOR WORK(3), IF THE
C                  STABILITY CHECK IS NEGATIVE, DEFAULT 0.5.
C
C      WORK(4), WORK(5)  PARAMETERS FOR STEP SIZE SELECTION
C                  THE NEW STEP SIZE FOR THE J-TH DIAGONAL ENTRY IS
C                  CHOSEN SUBJECT TO THE RESTRICTION
C                      FACMIN/WORK(5) <= HNEW(J)/HOLD <= 1/FACMIN
C                  WHERE FACMIN=WORK(4)**(1/(2*J-1))
C                  DEFAULT VALUES: WORK(4)=0.02D0, WORK(5)=4.D0
C
C      WORK(6), WORK(7)  PARAMETERS FOR THE ORDER SELECTION
C                  STEP SIZE IS DECREASED IF W(K-1) <= W(K)*WORK(6)
C                  STEP SIZE IS INCREASED IF W(K) <= W(K-1)*WORK(7)
C                  DEFAULT VALUES: WORK(6)=0.8D0, WORK(7)=0.9D0
C
C      WORK(8), WORK(9)  SAFETY FACTORS FOR STEP CONTROL ALGORITHM
C                  HNEW=H*WORK(9)*(WORK(8)*TOL/ERR)**(1/(J-1))
C                  DEFAULT VALUES: WORK(8)=0.65D0,
C                  WORK(9)=0.94D0 IF "HOPE FOR CONVERGENCE"
C                  WORK(9)=0.90D0 IF "NO HOPE FOR CONVERGENCE"
C
C      .....
C      IWORK(2)    THE MAXIMUM NUMBER OF COLUMNS IN THE EXTRAPOLATION
C                  TABLE. THE DEFAULT VALUE (FOR IWORK(2)=0) IS 9.
C                  IF IWORK(2).NE.0 THEN IWORK(2) SHOULD BE .GE.3.
C
C      IWORK(3)    SWITCH FOR THE STEP SIZE SEQUENCE (EVEN NUMBERS ONLY)
C                  IF IWORK(3).EQ.1 THEN 2,4,6,8,10,12,14,16,...
C                  IF IWORK(3).EQ.2 THEN 2,4,8,12,16,20,24,28,...
C                  IF IWORK(3).EQ.3 THEN 2,4,6,8,12,16,24,32,...
C                  IF IWORK(3).EQ.4 THEN 2,6,10,14,18,22,26,30,...
C                  IF IWORK(3).EQ.5 THEN 4,8,12,16,20,24,28,32,...

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C          THE DEFAULT VALUE IS IWORK(3)=1 IF IOUT.LE.1;
C          THE DEFAULT VALUE IS IWORK(3)=4 IF IOUT.GE.2.
C
C IWORK(4)  STABILITY CHECK IS ACTIVATED AT MOST IWORK(4) TIMES IN
C           ONE LINE OF THE EXTRAP. TABLE, DEFAULT IWORK(4)=1.
C
C IWORK(5)  STABILITY CHECK IS ACTIVATED ONLY IN THE LINES
C           1 TO IWORK(5) OF THE EXTRAP. TABLE, DEFAULT IWORK(5)=1.
C
C IWORK(6)  IF IWORK(6)=0 ERROR ESTIMATOR IN THE DENSE
C           OUTPUT FORMULA IS ACTIVATED. IT CAN BE SUPPRESSED
C           BY PUTTING IWORK(6)=1.
C           DEFAULT IWORK(6)=0 (IF IOUT.GE.2).
C
C IWORK(7)  DETERMINES THE DEGREE OF INTERPOLATION FORMULA
C           MU = 2 * KAPPA - IWORK(7) + 1
C           IWORK(7) SHOULD LIE BETWEEN 1 AND 6
C           DEFAULT IWORK(7)=4 (IF IWORK(7)=0).
C
C IWORK(8)  = NRDENS = NUMBER OF COMPONENTS, FOR WHICH DENSE OUTPUT
C           IS REQUIRED
C
C IWORK(10),...,IWORK(NRDENS+9) INDICATE THE COMPONENTS, FOR WHICH
C           DENSE OUTPUT IS REQUIRED
C
C .....
C
C IDID      REPORTS ON SUCCESSFULNESS UPON RETURN:
C           IDID=1  COMPUTATION SUCCESSFUL,
C           IDID=-1 COMPUTATION UNSUCCESSFUL.
C
C .....

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Subroutine ODEX2

Extrapolation code for second order differential equations $y'' = f(x, y)$ (Section II.14). It uses variable order and variable step sizes and is provided with a high-order dense output. The missing parts of the description are identical to those of ODEX.

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          SUBROUTINE ODEX2(N,FCN,X,Y,YP,XEND,H,
+              RTOL,ATOL,ITOL,
+              SOLOUT,IOUT,
+              WORK,LWORK,IWORK,LIWORK,LRCONT,LICONT,IDID)
C -----
C   NUMERICAL SOLUTION OF A SYSTEM OF SECOND ORDER
C   ORDINARY DIFFERENTIAL EQUATIONS  Y''=F(X,Y).
C   THIS IS AN EXTRAPOLATION-ALGORITHM, BASED ON
C   THE STOERMER RULE (WITH STEPSIZE CONTROL
C   ORDER SELECTION AND DENSE OUTPUT).
C .....
C
C   VERSION MARCH 30, 1992
C .....
C
C   Y(N)          INITIAL VALUES FOR Y
C

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